

# MUHAMMAD QASIM

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## EDUCATION

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### Masters

Computer Science, *University of Kentucky*, KY, USA

Aug. 2023 - May 2026

- Computer Vision - Medical Imaging - Intelligent Bioinformatics - Simulation Studies RNASeq

### UGRAD (Exchange Semester)

Computer Science, *Fayetteville State University*, NC, USA

Jan. 2022 - May 2022

- GPA: **4.0**, mentioned in the Chancellors List.

### BS (Hons)

Computer Science, *University of Engineering and Technology Lahore*

Oct. 2018 - Jan. 2023

- CGPA: **3.817**, among the top 3 students of the department. Expected first class Honors degree.

### High School (Intermediate)

(HSSC) Computer Science, *Punjab College for Boys Sialkot*

2016 - 2018

Gold Medalist, **94.09%** (1035/1100) with the first position in the whole division.

## TEACHING EXPERIENCES

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### Teaching Assistant - CS Department University of Kentucky, KY

Fall'23, Spring'24, Spring'26

I worked as a graduate teaching assistant for Software Engineering Lab CS216 and Introduction of Programming CS215 Lab and taught nearly 250 students and was ranked as one of the best TA's at UKY.

### External Examiner - CS Department UET Lahore

26 Jan 2023

Served as an external examiner for undergraduate final year project competition 2023.

### Teaching Assistant - CS Department UET Lahore

Fall'22

I worked as an undergraduate Teaching Assistant for Data Structures and Algorithms Lab in Fall 2022 semester.

### Online Machine Learning Training Workshops

Dec 2020 - Ongoing

I taught machine learning with hands on practice on real datasets in several training workshops.

## FIELD SKILLS AND INTERESTS

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### Courses

I have nearly two years of experience teaching beginner to intermediate level programming, Python and C++, analysis of algorithm, data structures and object-oriented programming as a teaching assistant at the University of Kentucky and University of Engineering and Technology.

### Programming Languages and Development Technologies

Python, R, C, C++, C#, SQL, Shell Scripting, OpenCV, NumPy, ScanPy, Scikit Learn, Pandas, Matplotlib, Keras, TensorFlow, HTML, CSS, Git, GitHub, Figma, OOP, Data Structures, Analysis of Algorithms, Parallel and Distributed Computing, High performance Cluster Computing.

## PUBLICATIONS

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### A Controlled Simulation-Based Benchmark for Transposable Element (TE) Quantification

May 25 - Ongoing

This study included an extensive workflow creation and analysis of TE quantification tools under a controlled simulation environment, and the downstream analysis to rank the quantification results. The process included bulk RNASeq reads simulation using polyester, varying the read length, library and the varying gene and TE combination. The plan is to create a unified NextFlow pipeline for TE quantification providing users with access to all state-of-the-art quantification tools under one workflow. [**Paper**] [*InProgress*]

### HERV Links with Cancer Patient Prognosis and Disease Status

May 24 - Oct 24

Systematic characterization of HERV at Pan-Cancer scale that can reveal shared and unique HERV patterns among different cancer types. Identify HERV signatures associated with prognosis, immune microenvironment to facilitate novel biomarkers and therapeutic target development. [**ORIEN 2024**] [*Poster*]

### The Effect of Variation of Time Interval of Denosumab (Prolia) Dosing on Biomarkers

Feb 24 - May 24

The study explored the effect of the varying time interval between Denosumab dosage varying on age, gender, bone density and the severity of Osteoporosis, with can be used as a potential biomarker for therapeutic target for patients with Osteoporosis. [**ASBMR 2024**] [*Abstract*]

## RESEARCH EXPERIENCE

### Research Assistant - Markey Cancer Center

May 24 - May 25

Working as a research assistant at MCC on RNASeq technologies for pan-cancer analysis and how presence of certain genes can be used as the biomarker for therapeutic target development for precision treatment.

### Leveraging Self-Supervised Learning for Medical Image Classification

Aug 24 - Dec 24

Utilizing advanced vision and Deep Learning based data-augmentation techniques to deal with data scarcity in medical imaging domain specifically Melanoma while keeping racial, data biases in the consideration, for effective non-invasive classification of Melanoma.

### Social Interaction Guide for Visually Impaired People

Jul 21 - May 22

Developing prototype mobile compatible machine learning and image processing models to be run on low power processors, integrate to an assistant application to enable visually impaired people to interact with the environment.

## WORK EXPERIENCE

### Ibex Global - Analyst Software Engineer

Nov 2022 - Jul 2023

Worked in R&D team to create enterprise level intelligent software. I worked on "Ibex Sense"; a computer vision powered enterprise solution to monitor customer service representatives' activities working for various global service campaigns.

### AxcelerateAI - Machine Learning Engineer

Oct 2022 - Nov 2022

Worked in R&D team of a tech startup to provide computer vision and machine learning solutions. I worked on English text-to-speech conversion.

### OmdenaAI - Junior Machine Learning Engineer (Remote)

Aug 2021 - Ongoing

It is a volunteer and availability-based position. The main job is data collection and processing for deep learning model training.

## INTERNSHIPS

### Internship Experience UK

June 2022

A 3 day long virtual internship arranged by Bright Network UK, where I learned and administrated effective leadership, self-motivated workflows and public policy making.

### Microsoft Learn Student Ambassador

2020 - 2022

Alpha level ambassador to train aspirants about Microsoft tools and services including azure, office 365, etc.

### Open-Source Contribution

2019 - Present

Developing programs and software, making it publicly available on GitHub and contribution to other's code.

## CERTIFICATIONS (MOOCs)

### Project Management Certification - Google - Coursera.org

Ongoing

### Deep Learning Specialization - Andrew NG - Coursera.org

Ongoing

### Machine Learning with Python - Cognitiveclass.ai

Completed

### Data Analysis with Python - Cognitiveclass.ai

Completed

## CONFERENCES AND WORKSHOPS

### ORIEN Data Science Summit - TPA

Oct 2024

I presented my work HERV Links with Cancer Patient Prognosis and Disease Status at Aster Insights ORIEN Data Science Summit at Tampa Florida, ranked among top 4 research posters.

### Open Data Science Conference

April 2022

Attended a 3 day long virtual conference about cutting-edge machine learning and data science technologies arranged by team ODSC Boston, MA.

### Getting Started with Artificial Intelligence

May 2020

Presented at an online training workshop arranged by IET on-campus UET Lahore, where the roadmap of machine learning with a practical programming implementation was taught.

## FIELD PROJECTS

### Pneumonia Detection Using Chest X-Ray

Developed a CNN based model to detect pneumonia using chest X-Ray images.

### Brain Tumor Detection Using MRI Images

Classification of malignant and benign brain tumor using MRI images and CNN based architecture.

### Image Classification on Intel Image Recognition Dataset

Deep CNN model trained on 24000 images with 6 natural scene classes and 89.54% accuracy.

### Traffic Sign Detection and Classification on GTRSB

Image classification CNN model trained on German Traffic Signs with 43 classes and 99.63% accuracy.

### Online Health Services Mobile App Prototype

Ideation, articulation, and development of doctor appointment booking application prototype using Figma.

### Portfolio Website

Designed my personal portfolio website using HTML, CSS, and JS to showcase the work done. [\[Link\]](#)

## HONORS AND AWARDS

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|-------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Exceptional Leadership Award - ADC</b>                                                                                                       | 2024 |
| Awarded the leadership award by ADC - Department of Computer Science for planning, organizing and hosting graduate students and faculty events. |      |
| <b>Chancellor's List - Fayetteville State University (NC, USA)</b>                                                                              | 2022 |
| In recognition of Scholastic Achievement for the Spring 2022 exchange semester.                                                                 |      |
| <b>Exchange Alumni at US Dept. of State</b>                                                                                                     | 2022 |
| Member of US exchange alumni network after successful completion of UGRAD exchange program.                                                     |      |
| <b>Global UGRAD Pakistan Participant - US Dept. of State</b>                                                                                    | 2021 |
| Winner of the highly competitive exchange program to the US, one of 137 selected from 16000 applicants.                                         |      |
| <b>Dean's List – University of Engineering and Technology Lahore</b>                                                                            | 2021 |
| Mentioned 6 times in recognition of educational excellence in 6 semesters by securing more than 3.7 GPA.                                        |      |
| <b>M.H Panhwar Talent Award</b>                                                                                                                 | 2018 |
| Awarded by Fundamental Rights Commission Pakistan as a tribute for scoring Gold Medal in HSSC-II.                                               |      |
| <b>Certificate of Merit - Govt. of Punjab</b>                                                                                                   | 2018 |
| Duly signed and granted by Chief Minister of Punjab on account of securing Gold Medal in HSSC-II.                                               |      |
| <b>Distinction Certificate - BISE Gujranwala</b>                                                                                                | 2018 |
| In the admiration of efforts to achieve first position and Gold Medal in HSSC-II.                                                               |      |
| <b>Gold Medal - BISE Gujranwala</b>                                                                                                             | 2018 |
| In the honor of securing first position in whole BISE Gujranwala annual HSSC-II exams.                                                          |      |

## STUDENT LEADERSHIP

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Graduate and Professional Senator - Student Government Association (SGA)</b>                                                                                                                                                                | Sep 2025 - Ongoing  |
| I always wanted to be a voice for a better opportunity for international students, so I joined Student Government to advocate the problems international graduate students face and to be able to provide suitable remedies to their concerns. |                     |
| <b>General Secretary - Association for Data and Computation (ADC)</b>                                                                                                                                                                          | Jan 2024 - May 2025 |
| Serving as GS for ADC at University of Kentucky. ADC is the only representative body for the graduate students in computer science, data science and computer engineering at UKY.                                                              |                     |
| <b>Secretary of IT - Youth Parliament Punjab</b>                                                                                                                                                                                               | 2021 - 2023         |
| Youth parliament is focused on leadership skills, critical mindset and meaningfully engaging youth in democratic decision making. The Ministry of IT is focused on technologically equipping youth with digital skills.                        |                     |
| <b>Campus Director - Hult Prize Global</b>                                                                                                                                                                                                     | 2021 - 2022         |
| Led the on-campus chapter of International social entrepreneurship organization that crowd sources ideas to solve the pressing problems of this world and arranged on campus entrepreneurship competition.                                     |                     |
| <b>Board of Governor - Velocity Productions</b>                                                                                                                                                                                                | 2021 - 2022         |
| As a student leader and Board of Governor, short films, and creative videography were co-directed by me.                                                                                                                                       |                     |
| <b>Head of Computer Science Chapter - IET on Campus UET Lahore</b>                                                                                                                                                                             | 2020 - 2021         |
| The computer science chapter of the IET was led by me where workshops and trainings sessions were arranged to teach the modern development technologies including machine learning and web development.                                        |                     |
| <b>Boxing and Martial Arts Player - University Official Team</b>                                                                                                                                                                               | 2019 - 2021         |
| Participated in National level martial arts championship (Bronze) and boxing championship (Qualifier).                                                                                                                                         |                     |

## CERTIFICATES

**Certificate of Appreciation:** IK Associates, ACCA Pakistan for securing first position in HSSC-II.

**Certificate of Participation:** High Achievers Development Workshop 2018, IOT Workshop 2021.

**Certificate of Completion:** (Online) Hour of Code, Data Analysis with Python, Introduction to Machine Learning.

## MY NON-ACADEMIC SELF

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I am a student entrepreneur; I have presented and won several cash prizes on my startups. Most recently, my startup called LOTWAYS, that uses an intelligent robot to autonomously manage parking spots, won the 1<sup>st</sup> prize in the Entrepreneurship Pitch Competition hosted by the University of Kentucky, entrepreneurship club [\[Link\]](#). Additionally, Equity, Diversity, and Inclusion are the areas of interest for me, I help students to find international educational scholarships, guide them about the requirements and application process. Boxing and MMA are my favorite sports. I have started to play pickleball recently. I am learning new languages Spanish most recently. Also, I have been an active member of the Developers Student Club, and Microsoft Learn Student Ambassador program.

## PORTFOLIO

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